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COMMENT

Commentary on "Musical harmonization with constraints: a survey"

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Abstract
Our 2001 survey on automatic harmonization appeared at a moment when tonal music offered an unusually clear test bed for constraint programming: explicit rule systems, finite domains, and musically meaningful combinatorics. Its citation activity turned out to be longer than we expected, even though we were neither the first nor the last to study the problem. Its most durable insight was the distinction between rule satisfaction and musical interest. Subsequent research confirmed that generating harmonizations that are legal is only part of the problem; representation, ranking, interaction, and style are at least as important. We also suggest that the role of constraints has evolved, from being the whole engine of a harmonizer to becoming a language of control inside hybrid symbolic, statistical, and neural systems.

Keywords Constraint programming · Music generation · Harmonization · Computational creativity · Controllable generation

1 Introduction

"Musical harmonization with constraints: A survey" reviewed a line of work at the intersection of tonal harmony, symbolic music representation, and constraint satisfaction [1]. The paper discussed expert systems, constraint logic programming, and consistency-based approaches to automatic four-part harmonization, and related them to our own attempts to structure musical objects and chord variables in a way that made search tractable [2–4]. Its closing claim was double-edged. On the one hand, it suggested that a restricted family of four-part harmonization problems could be handled effectively with constraint satisfaction

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techniques. On the other hand, it insisted that the larger musical question remained open, because syntactic correctness is not the same thing as musical interest [1].

What surprises us, in retrospect, is the afterlife of the paper. We were neither the first nor the last to work on harmonization, and yet the paper went on to be cited far more than we expected. A possible reason is that it named a problem that still has not disappeared. The most durable question is not whether harmonic rules can be encoded, they can, but how rule satisfaction, style, interaction, and evaluation should be combined in systems meant for actual musicians. This commentary revisits what the paper captured well, what it overstated, and what later work, including our own, helped clarify.

2 From a survey to a research program

The paper condensed several years of work in which the same question kept returning in different forms: why does a task that looks naturally declarative become difficult as soon as one seeks complete, musically legal solutions? Earlier papers had already shown that the choice of representation was at least as important as the solving power of the engine [2, 3]. Roy's thesis made this point explicit by organizing harmonization problems around structured musical objects and intermediate chord descriptions rather than flat note tuples [4].

Our survey also stood in an existing lineage. The idea of automatic harmonization as a formal knowledge-engineering problem was already embodied in Kemal Ebcioğlu's CHORAL work, which became a landmark reference in computer music [5]. Philippe Ballesta's doctoral work then showed how these questions could be addressed in a structured constraint-based setting, using Ilog's PECOS system, described in our survey as an ancestor of Ilog Solver [1, 6]. Our paper was an attempt to take stock of a field that had already accumulated several distinct technical traditions.

Harmonization was an attractive domain for constraint programming for obvious reasons. Tonal music comes with explicit rule sets, finite domains, and several interacting temporal scales. At the same time, it is not a toy problem. The moment one moves from isolated chord legality to full exercises with fixed voices, hidden chord functions, and voice-leading requirements, one encounters the issues that make modeling matter: the interaction between local and non-local constraints, the importance of decomposition, and the gap between a valid solution and a convincing one [1].

With hindsight, our 2001 paper could be seen as a research program. It argued implicitly that artistic applications deserve serious constraint models, and that they can feed back into the development of modeling techniques themselves.

3 What the paper got right?

One thing the paper got right was to insist that harmonization is naturally constraint-based. This may sound obvious now, but at the time it mattered to show that musical knowledge could be expressed as explicit relations over notes, chords, and short temporal neighborhoods without reducing music to a mere list of prohibitions [1]. Later surveys and systems confirmed that constraint programming remained a powerful way to encode music theories

beyond the Bach-chorale setting, including richer harmonic, rhythmic, and formal domains [7].

A second point that aged well is the importance of representation. Our own contribution in the survey was to structure the problem so that chord information could mediate between note-level choices and longer-range harmonic regularities [1, 2, 4]. This lesson remains central. In creative domains, performance rarely depends only on the solver. It depends on which variables one chooses, at what musical level decisions are taken, and which intermediate abstractions are made explicit.

The third point, maybe the most important, was the distinction between syntactic correctness and musical interest. The conclusion of the 2001 paper was that the technical problem of four-voice harmonization could be considered largely solved, but that the problem of producing "musically nice or interesting melodies" remained open [1]. That diagnosis still seems correct. If anything, later progress in statistical and neural generation made the point even clearer: feasibility is not enough, and style is not enough either. Users need systems that can be guided, revised, and steered.

4 What we would revisit today

The sentence about the problem being "solved" is the one we would revisit more carefully today. For a restricted family of exercises, fixed style, explicit rule base, finite horizon, and a representation designed for the task, the statement was defensible [1]. Outside these boundaries, it was too strong. Practical harmonization involves ranking, stylistic adaptation, partial specifications, interaction, and often real-time constraints. These dimensions change the nature of the problem as much as the size of the search space.

As of 2026, the problem still motivates new work. Recent papers by Damien Sprockeels and Peter Van Roy revisit tonal harmony with modern constraint-programming tools, first through a model of diatonic four-voice harmony and then through the broader Harmoniser project for assisting composers [8, 9]. Each generation of researchers reformulates the problem with new modeling tools, new expectations of interactivity, and a different balance between formalization and usability.

We would also place more emphasis on evaluation. In the original paper, evaluation was primarily framed in terms of legality, coverage of the rule set, and computational efficiency. Those criteria were appropriate at the time, but they would be considered incomplete now. A useful harmonization system must also support user intent, expose musically meaningful controls, and produce a diversity of acceptable outputs rather than a single canonical answer. In other words, the real question is not only whether a solver can find a harmonization, but whether a musician can work with the system.

Finally, we would state more clearly that the paper was intentionally narrow in musical scope. Tonal four-part writing was an excellent domain for investigation because its theory is highly formalized, not because it exhausts what musical intelligence is. The success of the constraint-based approach in this setting should therefore be read as a proof of relevance, not as a universal model of composition.

5 From solving to steering

Since 2001 the role of constraints has changed. In many later systems, constraints are no longer the entire generator; instead, they provide the language of control that makes generation usable. This evolution was already visible in interactive style systems such as the Continuator, which shifted the focus from rule satisfaction to responsive musical dialogue [10]. It became explicit in work on steerable sequence generation, where constraint programming was used to recover user control over probabilistic models [11].

This line of work led naturally to hybrid approaches. In non-conformant harmonization, the challenge was no longer merely to satisfy Baroque rules, but to preserve hard musical requirements while borrowing stylistic material from corpora that did not originally contain the target song [12]. Around the same period, DeepBach marked a different move: the rules were no longer given explicitly, but learned from data, while still allowing (positional) user-imposed constraints and interactive editing [13]. Anticipation-RNN pushed the same intuition further by building the handling of user constraints directly into a neural generation architecture [14].

Seen from today, the most important continuity is the persistence of a musical requirement that the 2001 paper already made visible: composition is rarely a purely left-to-right process, and creative tools must accommodate partial decisions, revisions, and future-oriented constraints. In that sense, the central question has shifted from solving to steering.

6 Lessons for constraint programming and creative AI

The first lesson is that constraints remain valuable when they express intent rather than mere correctness. Musicians do not usually think in terms of complete solutions; they think in terms of partial commitments, local obligations, stylistic expectations, and editable choices. Constraint-based models are well suited to represent such situations because they let users specify what must hold while leaving room for search or sampling elsewhere.

A related lesson concerns the dynamics of modeling itself. In problems like harmonization, one usually starts from the right intuition: the task seems to lend itself naturally to hard, well-defined properties. This intuition is often correct at the beginning. But as the model evolves, one is tempted to add more and more properties to make the results look, or here sound, better. At that point, a phenomenon appears, that we have observed repeatedly in practice: solvers seem to have a natural tendency to find precisely those annoying solutions first. In other words, they exploit the "holes" in the model before they reveal the musically convincing region one had in mind. A generated harmonization may, for instance, contain long series of repeated notes that violate no explicit rule yet still sound degenerate. We suspect that this tendency is not accidental, but reflects something deeper about modeling itself: reality is always richer than the properties we manage to state explicitly, so the first exact solutions of a formal model often expose its blind spots more readily than its intended meaning. The natural reaction is to patch the model, first with strong constraints such as *AllDiff*, then with softer or more global distinct-value constraints that we later explored in playlist generation [15]. As we wrote in our later commentary on FlowComposer, even in the playlist case we started from the intuition that good playlists "should satisfy various contradictory criteria, dealing both with continuity and variety" [16]. That line of work

led to the introduction of the NValue constraint, but also illustrated a broader point: added properties do remove some degenerate solutions, yet they also change the meaning of the model and increase its complexity. After a while, adding one more constraint either becomes intractable or simply misses the point, because the issue is no longer constraint satisfaction proper, but the difficulty of formalizing musical value itself.

The second lesson is that creative applications expose a limitation of the classical success criterion in combinatorial problem solving. In harmonization, as in many artistic tasks, there is rarely a best solution and often no agreed objective function. This is precisely why the domain is interesting. It forces us to ask how constraint satisfaction should interact with optimization, probabilistic modeling, interface design, and human judgment.

The third lesson is that there seems to be a practical limit to our ability to elicit all the relevant constraints explicitly. Rules of harmony can be stated, refined, and implemented with great precision, but at some point the knowledge-acquisition bottleneck becomes the real obstacle. This is one reason why learned models became attractive, because much of musical competence remained difficult to enumerate in symbolic form.

The fourth lesson is that hybridization should not be seen as a retreat from symbolic modeling. On the contrary, later work suggests that learned models and constraint-based control are complementary. Learning is powerful for capturing style, regularity, and variation from corpora; constraints are powerful for editing, guaranteeing, and communicating structure. The most promising systems are often those that let these two modes of description cooperate rather than compete [11, 13, 14].

Finally, the distinction between technically correct and interesting music has only become sharper. We can now generate much more technically plausible material than in 2001, as illustrated by recent research systems such as MusicLM and MusicGen, and by commercial platforms such as Suno [17–19]. But the next step, generating music that is consistently interesting, moving, or simply worth keeping, still does not seem reachable through explicit knowledge alone. This reinforces the intuition of our original conclusion: the hard part is not legality, but musical value.

7 Conclusion

Harmonization was, and still is, a useful research problem because it reveals the difference between validity and musicality. That difference has shaped much of the work that followed. It is visible in the move toward interactivity, in the search for controllable statistical and neural models, and in the renewed importance of representation.

For the history of *Constraints*, the broader message is that artistic applications are not marginal showcases for existing techniques. They are demanding test beds that reveal that modeling languages are not for only finding solutions, but for helping users formulate, revise, and explore a space of possibilities. In that respect, harmonization has perhaps been more fertile as a trigger for research ideas than as a source of indispensable musical tools.

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